INCH-POUND

A-A-54721
03 August 1992

COMMERCIAL ITEM DESCRIPTION

WATER BATH, ELECTRIC

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-W-37090 which is cancelled.

This Commercial Item Description covers an electrically heated water bath with a gable type cover, suitable for use in laboratory testing.

Salient characteristics:

Design and construction. The water bath shall be operable from 115 volt, 50/60 Hz, AC 1000 watt supply lines. The water bath housing shall be double walled with insulation between walls, including the bottom. The insulation shall be non-hygroscopic, non-packing, non-powdering material with thermal conductivity coefficient not exceeding 0.27 BTU/HR/SQFT/^{O}F/inch thickness. The exterior walls shall be of corrosion-resisting steel .025 inch thick No. 24 United States Standard (USS) Gage. Exterior nominal dimensions shall be: Length 15 inches, width 14 inches, and height 17 inches. The nominal water chamber dimensions shall be: Length 13 inches, width 12 inches and depth 7 inches. Shall be designed so that the water chamber can accommodate six Wassermann-Type laboratory test tube racks that can hold twenty 13 mm test tubes per rack, and each rack having overall dimensions as follows: Length 10 inches, width 2-1/16 inches and height 4 inches. Note that the laboratory test tube racks are not required to be supplied with the water bath. The bottom and sides of the water chamber shall be made of corrosion-resisting steel nominally .020 inches thick. The water chamber (inside) and the outside body shall be of welded construction. The water chamber (inside) shall be soldered watertight.

AMSC N/A FSC 6640

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The water bath shall have a removable galvanized steel base .060 inches thick nominal. The bottom of the unit shall have protruding feet. The water bath shall contain the following components:

Cover. The cover shall be fabricated of corrosion-resisting steel and shall be the gable type. The cover shall have a plastic handle on the top and two cut-outs on one side. The cut-outs shall provide space for the thermoregulator and thermometer holder with thermometer. To prevent the cover from sliding off or into the water chamber, the bottom of the cover shall protrude into the water chamber and shall have a flange on the sides that rests on the top of the water bath. The design and dimensions shall be such that the cover can be easily set in place and can be easily removed.

Diffuser platform. Shall be an expanded metal platform to support all test apparatus and to protect the heating coils and stirring bar. The diffuser platform shall be made of corrosion-resisting steel. Shall be polished on both sides with a suitable commercial finish. All sides shall be bent in order to raise the platform 1-1/4 inches above the bottom of the water chamber.

Stirring Bar. Stirring shall be achieved with a magnetic bar sealed in molded Teflon, located in a magnetic field at the bottom center of the water chamber on a centering disk and surrounded by a flow director. Nominal length of the stirring bar shall be 2-1/2 inches.

Centering Disk. The centering disk for the stirring bar shall be of a suitable corrosion-resisting metal. The disk shall be approximately .020 inches thick and 3 inches in diameter.

Flow Director. The water chamber shall have a flow director, which surrounds the stirring bar, and shall be designed to achieve uniform circulation.

Heating element. Immersion type low wattage copper sheathed density heaters shall be installed on the bottom of the water chamber. The heater shall be rated 1000 watts at 120 volt.

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Stirring motor. The magnetic circulator shall be energized from below by an induction type, non-sparking motor. The motor shall operate at 120 volts, 50-60 Hz, 35 watts nominal. Shall drive a magnet below the water chamber and a fan below the motor. The fan shall draw hot air away from the motor, through vents in the sides and back of the water bath.

Thermoregulator. The thermoregulator shall be an immersible type, mercury sealed in glass, connected to an electronic relay switch assembly. The thermoregulator shall be graduated in 1°C increments with total range of -10° to +110°C. There shall be a means of setting the temperature at the top of the stem. The thermoregulator shall be secured in a corrosion-resisting metal bracket that shall be securely fastened to the water bath.

Thermometer. The thermometer shall be partially immersion mercury-in-glass type, and shall have nominal dimensions 10 inches long, and 1/4 inches in diameter. As a minimum the thermometer shall have a 0-100°C range with 1°C graduations. The thermometer shall have an accuracy of ± 0.5°C. The thermometer shall be secured in a corrosion-resisting steel holder.

Water level regulator and drain valve. The constant water level control and drain cock shall be made of nickel-plated brass 1/8 National Taper Pipe (N.T.P) size.

Control panel housing. The control panel housing shall be made of a suitable plastic and shall be located on the front of the water bath. The housing shall have a nameplate, heater switch and pilot light.

Nameplate. The nameplate shall be permanently and legibly marked with the following information: manufacturer's name or registered trademark, electrical characteristics and serial number.

Heater switch. The heater switch shall be an On-Off line type, single pole, 15 amp, 125 vac.

Pilot light. The pilot light shall light when the heating element is energized and shall not light when the heater is not energized.

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Motor switch. The motor switch shall be an On-Off line type, single pole, 15 amp, 125 vac.

Power cord. A three conductor power cord with molded plug shall be supplied with the water bath. A strain relief bushing shall be provided where the power cord enters the water bath unit. The plug shall be three bladed and be for use with receptacles shown in ANSI Standard C73.11 or shown in Configuration No. 5-15 of NEMA Standard WD 6-88.

Performance.

Dielectric strength. There shall be no breakdown when the water bath is subjected to 1240 vac applied between current carrying parts and metal enclosure.

Temperature stability. The indicated temperature shall be constant within $\pm$ 0.05 ^{o}C.

Operating temperature range. Water in the uncovered water bath shall reach a temperature of 60°C, when the thermoregulator is set at 60°C and the heating system turned on. Water in a covered water bath shall reach a temperature of 100°C when the thermoregulator is set at 100°C.

Heating time. The water bath shall reach a constant temperature of 100°C within 2-1/4 hours, when filled with water of 20°C, cover in place, and the thermoregulator set at 100°C and heating system turned on.

Workmanship. The water bath shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One water bath, as specified, constitutes one unit.

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QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Inspections and testing. Inspections and testing shall be conducted on the water baths in accordance with commercial practice. Sampling for the inspections and testing shall be conducted in accordance with commercial practice. In addition, each water bath shall be inspected and tested to demonstrate its capability for proper operation. Also, one water bath shall be selected and tested to determine the heating time in accordance with the following:

> The water bath, initially at room temperature, shall be filled with water at 20°C ± 1°C. The water bath shall then be covered and the heating system turned on with the thermoregulator set for 100°C. The time for the water bath to reach 100°C shall be monitored and recorded. A water bath that does not reach 100°C within 2-1/4 hours shall be cause for rejection.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Other publications. The following documents form a part of this specification to the extent that is specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C73.11, Plugs and Receptacles - General Purpose 125 Volts, 15 Amperes, 2 Poles, 3 Wire Grounding Type, Dimensions of

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

WD 6-88 Wiring Devices - Dimensional Requirements (Revision and Redesignation of ANSI C73 - 73)

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW Suite 300, Washington, DC 20037.)

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

NSN	Item Identification
6640-00-765-0599	WATER BATH, ELECTRIC, PRECISION

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MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6640-1762

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	A-A-54721	03 AUGUST 1992

3 DOCUMENT TITLE

WATER BATH, ELECTRIC

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)*

5 REASON FOR RECOMMENDATION

6. SUBMITTER

	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON	e. [illegible] (Optional)

8. PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code)	
Defense Personnel Support Center	(1) Commercial (215) 737-2870	(2) AUTOVON 444-2870
c ADDRESS (Include Zip Code) ATTN: MSE 2800 S. 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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